

Streamlining Ticket Assignment for Efficient Support Operations

Category: ServiceNow Application Developer

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Course/Batch: Computer Science and Engineering

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1. Introduction

Efficient support operations are essential for ensuring quick resolution of incidents and maintaining customer satisfaction. In manual ticket assignment, agents or administrators need to manually review tickets and assign them to the appropriate support group. This often causes delays, misrouting of issues, and a lack of accountability.

To overcome these challenges, ServiceNow's automation capabilities can be leveraged. By using Flow Designer, User & Group Management, and Access Control Lists (ACLs), we can design an automated ticket assignment process that ensures incidents are assigned to the right group in real-time.

2. Skills & Tools Used

ServiceNow Platform

- User and Group Management
- Flow Designer
- Access Control Lists (ACLs)

3. Problem Statement

Current manual ticket assignment delays resolution, leads to inefficiency and dissatisfaction. Need for automation.

Manual ticket assignment leads to:

- Delay in resolving incidents.
- Misallocation of tickets to the wrong groups.
- Increased dissatisfaction among users and support teams.

There is a need for an **automated ticket routing mechanism** that assigns tickets to the correct group without human intervention.

4. Proposed Solution

Automated ticket routing system in ServiceNow using Flow Designer. Proper ACLs to secure ticket data. Role-based access for support teams.

- ❑ Create **users, groups, and roles** in ServiceNow to represent the support structure.
- ❑ Design an **automated flow** in Flow Designer that routes tickets to groups based on ticket category.
- ❑ Implement **ACLs** to secure ticket data and ensure that only authorized users have access.
- ❑ Test the flows thoroughly with real-time scenarios to validate the automation

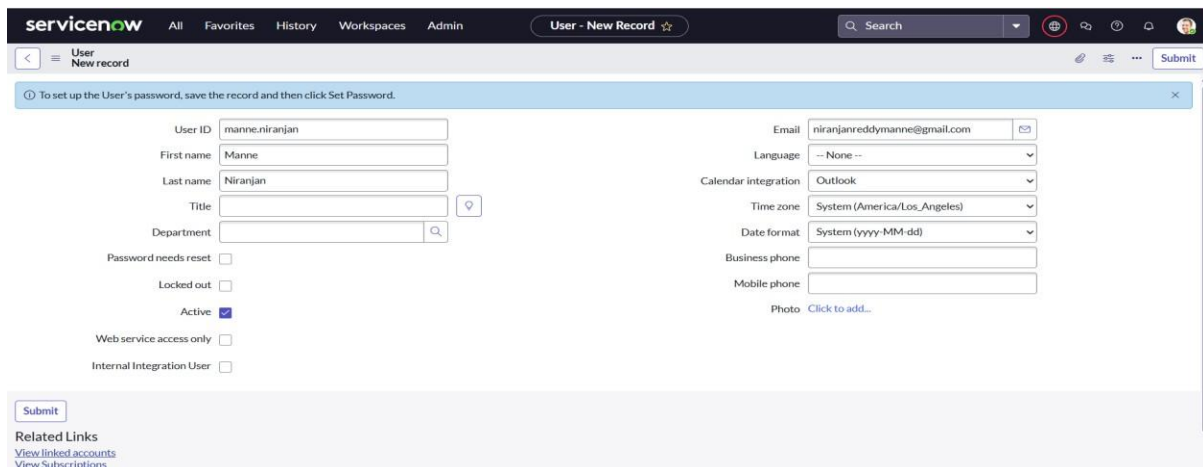
5. Implementation Steps

Step 1: User Creation

New users are created in ServiceNow to represent support agents

- Create new users in ServiceNow to represent support agents.
- Example:
 - **User 1**
 - **User 2**
- Each user will have unique login credentials and contact details in the system.

User 1:



The screenshot shows the 'User - New Record' form in ServiceNow. The form is divided into two main sections: 'User Information' on the left and 'Contact Information' on the right. The 'User Information' section includes fields for User ID (manne.niranjan), First name (Manne), Last name (Niranjan), Title, and Department. There are also checkboxes for 'Password needs reset', 'Locked out', 'Active' (checked), 'Web service access only', and 'Internal Integration User'. The 'Contact Information' section includes fields for Email (niranjanreddymanne@gmail.com), Language (None), Calendar integration (Outlook), Time zone (System (America/Los Angeles)), Date format (System (yyyy-MM-dd)), Business phone, and Mobile phone. A 'Photo' field with a 'Click to add...' link is also present. At the bottom left, there is a 'Submit' button and a 'Related Links' section with links to 'View linked accounts' and 'View Subscriptions'.

User 2:

The screenshot shows the ServiceNow 'User' form for Katherine Pierce. The form is divided into two main sections. The left section contains fields for User ID (Katherine.Pierce), First name (Katherine), Last name (Pierce), Title, and Department. Below these are checkboxes for 'Password needs reset', 'Locked out', 'Active' (checked), 'Web service access only', and 'Internal Integration User'. The right section contains fields for Email (katherine.pierce@gmail.com), Language (None), Calendar integration (Outlook), Time zone (System (America/Los Angeles)), Date format (System (yyyy-MM-dd)), Business phone, Mobile phone, and Photo (Click to add...). At the bottom of the form are buttons for 'Update', 'Set Password', and 'Delete'. Below the form, there are 'Related Links' for 'View linked accounts', 'View Subscriptions', and 'Reset a password'.

Step 2: Group Creation

Groups are defined to organize users into specialized teams for ticket handling.

Define groups to organize users into specialized support teams.

Example:

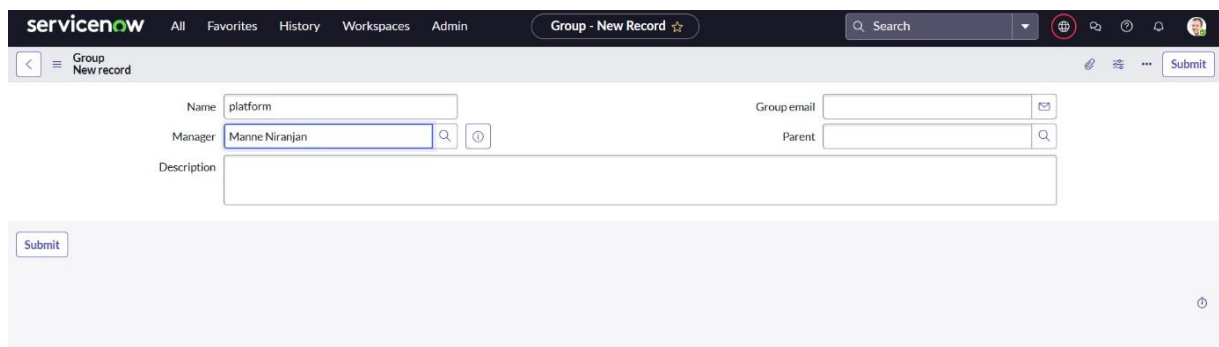
- **Group 1:** Operations Support Group (handles operations-related issues)
- **Group 2:** Platform Support Group (handles platform-related issues)

Each group can contain one or more users.

Group 1:

The screenshot shows the ServiceNow 'Group' form for 'certificates'. The form has fields for Name (certificates), Group email, Manager (Katherine Pierce), and Parent. Below these is a large text area for Description. At the bottom of the form are buttons for 'Update' and 'Delete'.

Group 2:

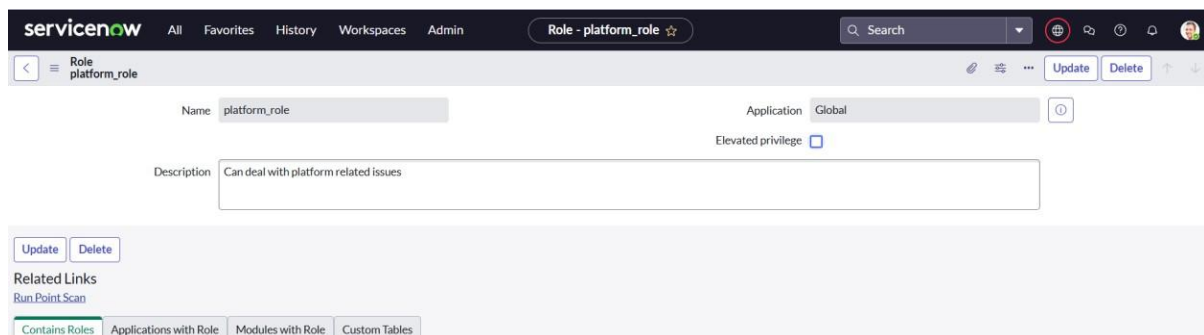


The screenshot shows the 'Group - New Record' form in ServiceNow. The form has a dark header with the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, Admin. A search bar is on the right. Below the header, there's a breadcrumb 'Group - New Record' and a 'Submit' button. The form fields include: Name (platform), Manager (Manne Niranjan), Group email, Parent, and Description. A 'Submit' button is at the bottom left.

Step 3: Role Creation

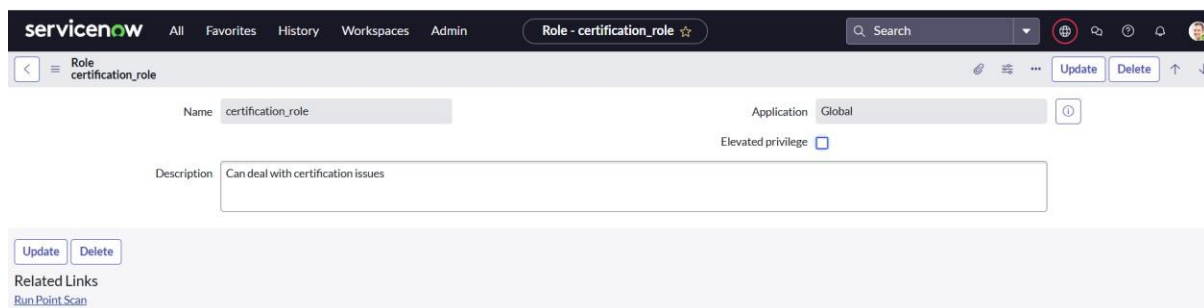
- Define roles to manage access and permissions.
- Example:
 - **Role 1:** Operations_Role – allows access to operations tickets.
 - **Role 2:** Platform_Role – allows access to platform-related tickets.

Role 1:



The screenshot shows the 'Role - platform_role' form in ServiceNow. The form has a dark header with the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, Admin. A search bar is on the right. Below the header, there's a breadcrumb 'Role - platform_role' and buttons for 'Update' and 'Delete'. The form fields include: Name (platform_role), Application (Global), Elevated privilege (checkbox), and Description (Can deal with platform related issues). Below the form, there are 'Update' and 'Delete' buttons, and a 'Related Links' section with a link to 'Run Point Scan'. At the bottom, there are tabs for 'Contains Roles', 'Applications with Role', 'Modules with Role', and 'Custom Tables'.

Role 2:



The screenshot shows the 'Role - certification_role' form in ServiceNow. The form has a dark header with the ServiceNow logo and navigation tabs: All, Favorites, History, Workspaces, Admin. A search bar is on the right. Below the header, there's a breadcrumb 'Role - certification_role' and buttons for 'Update' and 'Delete'. The form fields include: Name (certification_role), Application (Global), Elevated privilege (checkbox), and Description (Can deal with certification issues). Below the form, there are 'Update' and 'Delete' buttons, and a 'Related Links' section with a link to 'Run Point Scan'.

Step 4: Tables Creation

- Create custom tables to represent ticket-related data.

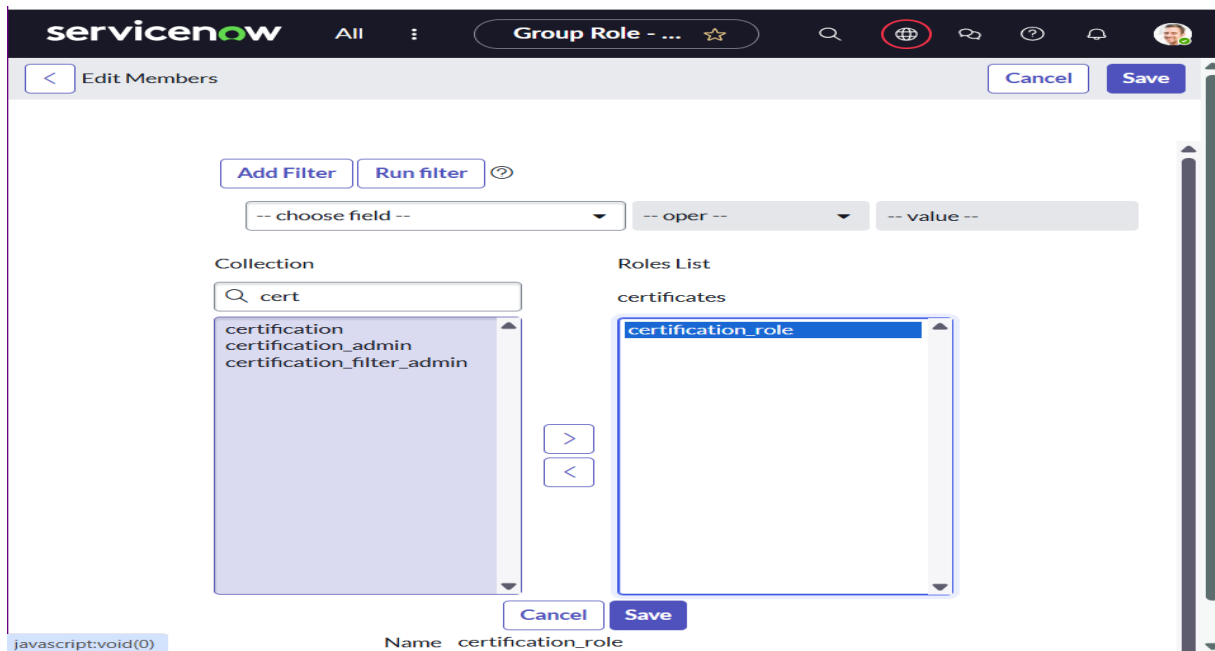
- Example: **Operations_Tickets Table** to log operational issues.

Table – Operations related

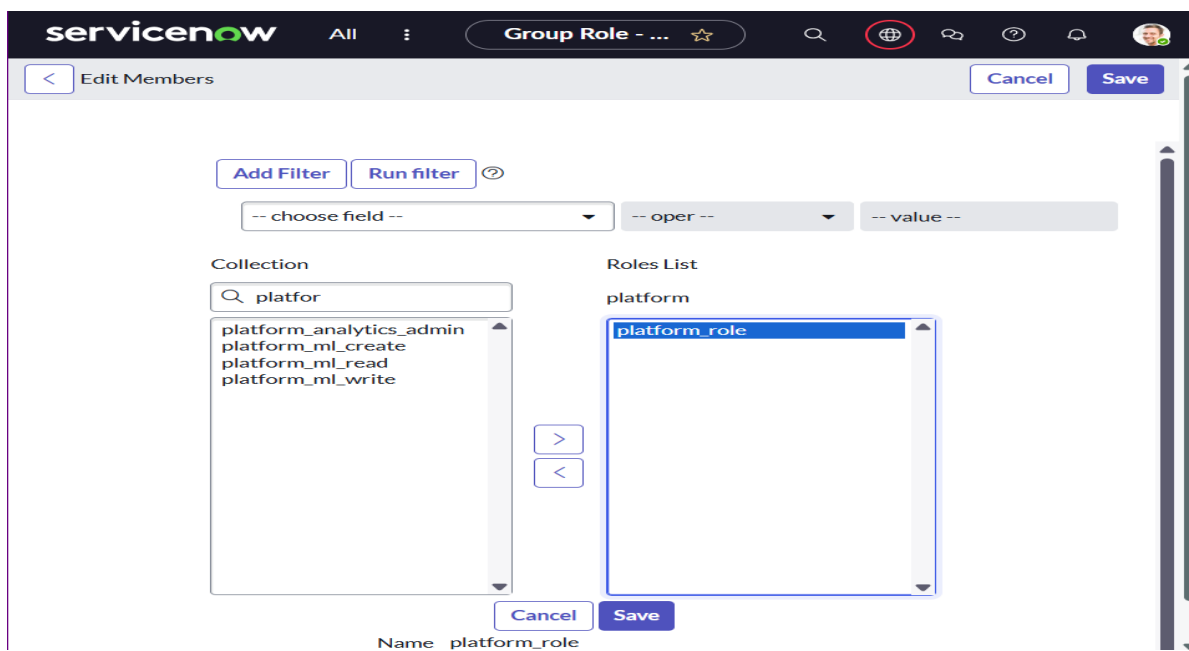
Step 5: Assigning roles to users & groups

- Assign roles to ensure users have the correct level of access.
- Example:
 - Assign **Operations_Role** to User 1 and Operations Support Group.
 - Assign **Platform_Role** to User 2 and Platform Support Group.

Assigning role and user to Certificate group



Assigning role and user to Platform group



Edit Members

Collection: manne

Group Members List: platform

Manne Niranjana

Name: Manne Niranjana
First name: Manne
Last name: Niranjana

Step 6: Assign role to table

☐ Ensure that only authorized roles can access the new tables.

☐ Example: Operations_Role can view/edit Operations_Tickets, while Platform_Role can view/edit Platform_Tickets.

Access Control u_operations_related

Definition

Access Control Rules allow access to the specified resource if *all three* of these checks evaluate to true:

1. The user has one of the roles specified in the **Role** list, or the list is empty.
2. Conditions in the **Condition** field evaluate to true, or conditions are empty.
3. The script in the **Script** field (advanced) evaluates to true, or sets the variable "answer" to true, or is empty.

The three checks are evaluated independently in the order displayed above.

[More Info](#)

Requires role

Role
✗ u_operations_related_user
✗ Platform_role
✗ Certification_role
+ Insert a new row...

1 to 3 of 3

Step 7: Creation of ACL

☐ Configure **Access Control Lists (ACLs)** to restrict data access.

☐ Example:

- Only Operations Support Group can read/write Operations_Tickets.

- Only Platform Support Group can access Platform_Tickets.

The screenshot shows the ServiceNow interface for configuring an Access Control rule. The rule is named "u_operations_related". The "Conditions" tab is active, showing a "Requires role" section with a table of roles: "u_operations_related_user", "platform_role", and "certification_role", each with a red "X" indicating it is required. Below this is a "Security Attribute Condition" section with radio buttons for "Existing" and "Local" (selected), a search field for "Security Attribute", and a "Condition" field.

Step 8: Create a Flow to Assign operations ticket to group

Step 9: Flow Creation – Operations Tickets

- Using **Flow Designer**, create a flow:
 - Trigger: When a new ticket is created with category = Operations.
 - Action: Automatically assign the ticket to **Operations Support Group**.

The screenshot shows the ServiceNow Workflow Studio interface for creating a new flow. The form is titled "Let's get the details for your flow". It includes fields for "Flow name" (set to "Regarding Certificate."), "Application" (set to "Global"), "Description" (set to "Describe your flow."), "Protection" (set to "-- None --"), "Run as" (set to "System user"), and "Flow priority default" (set to "Medium (default)"). There are "Cancel" and "Build flow" buttons at the bottom right.

TRIGGER

Operations related Created or Updated **Trigger: Created or Updated** (Regarding certificates)

Trigger: Created or Updated

* Table: Operations related [u_operations_related]

Condition: All of these conditions must be met

Issue is Regarding certificates

OR AND

New Criteria

Run Trigger: For every update

Advanced Options

Delete Cancel Done

Workflow Studio Regarding Certificate". Flow

Regarding Certificate". Inactive

TRIGGER

Operations related Created or Updated where (Issue is Regrading Certificates)

ACTIONS Select multiple

+ Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Step 10: Flow Creation – Platform Tickets

- Create another flow:
 - Trigger: When a new ticket is created with category = Platform.
 - Action: Automatically assign the ticket to **Platform Support Group**.

Step 11: Flow Activation

- Activate both flows in Flow Designer so that they run automatically in real-time.

Workflow Studio

Regarding Certificate*. Flow

Regarding Certificate*. Active

View: Test Deactivate Activate Save

TRIGGER

Operations related Created or Updated where (Issue is Regrading Certificates)

ACTIONS Select multiple

1 Update Operations related Record

Add an Action, Flow Logic, or Subflow

ERROR HANDLER

If an error occurs in your flow, the actions you add here will run.

Data Collapse All

Flow Variables

Trigger - Record Created or Updated

Operations related Record Record

Changed Fields Array.Object

Operations related Table Table

Run Start Time UTC Date/Time

Run Start Date/Time Date/Time

1 - Update Record

Operations related Record Record

Operations related Table Table

Action Status Object

servicenow

All

Access Contr...

Access Control u_operations_related

Type record

Application Global

Operation Active

read

Decision T

Allow If

Admin ove

Protection

-- None --

Name u_operati

Description

Default access control on u_operations_related

Applies To

No. of records matching the condition: 0

(empty)

Conditions

System Administra...

System Administrator

ServiceNow

Profile

Elevate role

Elevate a role by adding privileges, which end when you log out. [Learn more](#)

AVAILABLE ROLES

☒ security_admin

Grant modification access to High Security Settings, allow user to modify the Access Control List

Cancel Update

Step 12: Testing the Flow

Once the flow was designed, it was tested by creating sample incidents/tickets in ServiceNow. The testing ensured that tickets were automatically assigned to the correct groups based on the issue category

Steps in Testing

1. Create a Test Ticket

- Example: Create an incident with category '*Unable to login to platform*'.
- Expected Result: The ticket should be auto-assigned to the *Platform Support Group*.

2. Check Assignment

- Verify that the ticket was routed to the correct support group automatically.

Testing ensures the automation works correctly.

Test Case 1 – Operations Ticket

- Action: Create a ticket with category = "Database issue (Operations)".
- Expected Result: The system auto-assigns the ticket to **Operations Support Group**.
- Verification: Check the "Assigned Group" field → should show Operations Support Group.

Test Case 2 – Platform Ticket

- Action: Create a ticket with category = "Unable to login to Platform".
- Expected Result: The system auto-assigns the ticket to **Platform Support Group**.
- Verification: Check the "Assigned Group" field → should show Platform Support Group.

Test Case 3 – Unauthorized Access

- Action: Try accessing Platform_Tickets with a user who only has Operations_Role.
- Expected Result: Access denied due to ACLs.
- Verification: User should not be able to view/edit platform tickets.

6. Results

- Tickets were successfully routed to the correct groups based on their category.
- No manual intervention was required.
- Unauthorized users were unable to access restricted tickets due to ACLs.

7. Benefits

- Faster ticket resolution.
- Reduced manual workload.
- Improved customer and user satisfaction.
- Clear accountability since tickets are assigned to the right group instantly.

8. Conclusion

The automation of ticket assignment in ServiceNow using Flow Designer significantly improved efficiency in support operations. It reduced delays, ensured correct routing, and enhanced security through ACLs.

9. Future Enhancements

- Add **priority-based routing** (e.g., high-priority tickets assigned to senior staff).
- Use **Machine Learning (ML)** to predict the best group for complex issues.
- Integrate with **Slack/Email notifications** for real-time updates to support staff.
- Implement **auto-escalation rules** if tickets remain unresolved beyond SLA.